AC circuit

(i) Diode ***

(ii) Zener Diode

(iii) Rectifier 2 T

(iv) Transistor

BJT

3 Terminal

(i) Emitter

(ii) Base

(iii) Collector

$+V_C - V_B$

$= V_{CB}$

$+V_E - V_B$

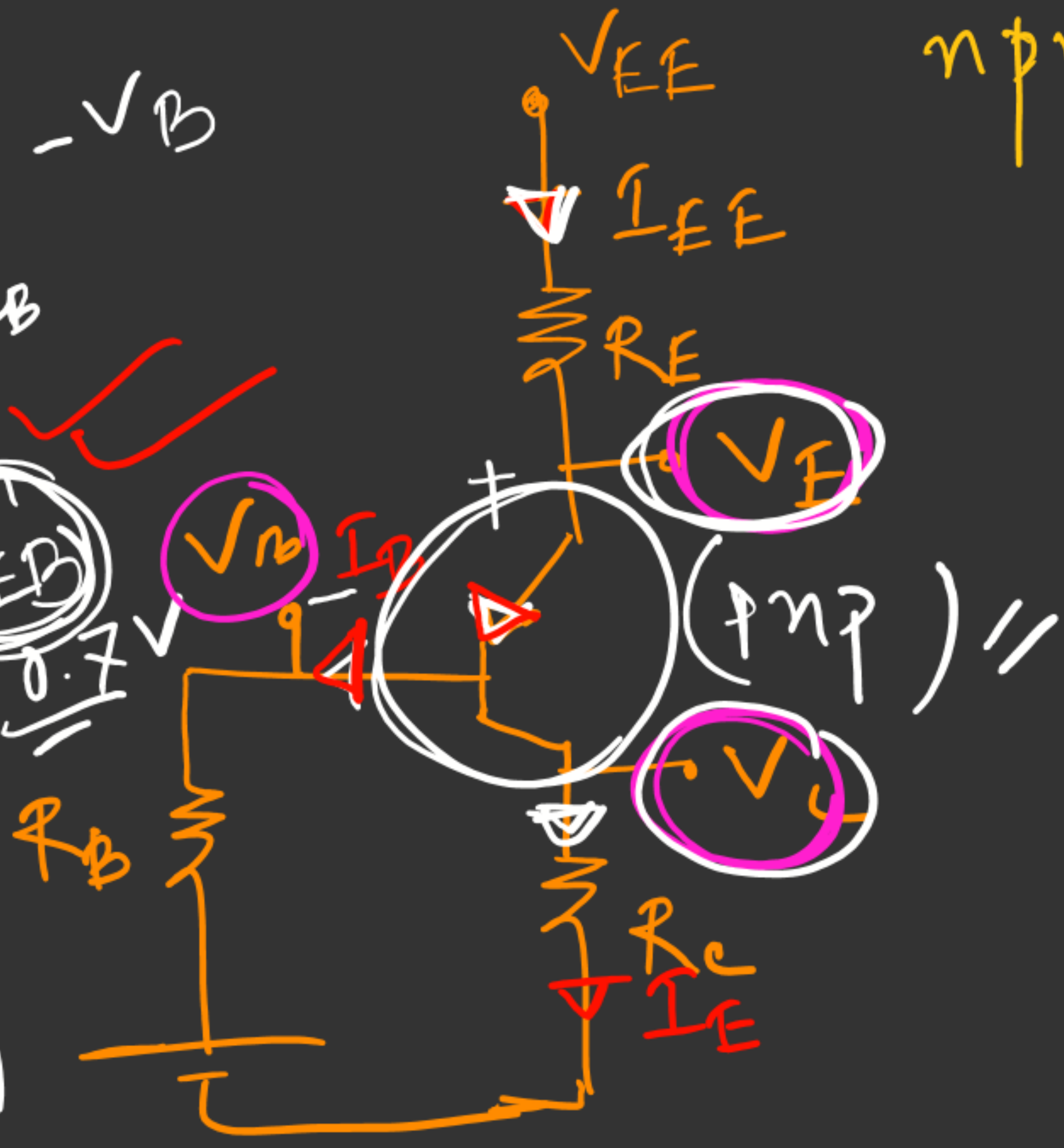
$= V_{EB}$

$= 0.7\,V$

$\downarrow$

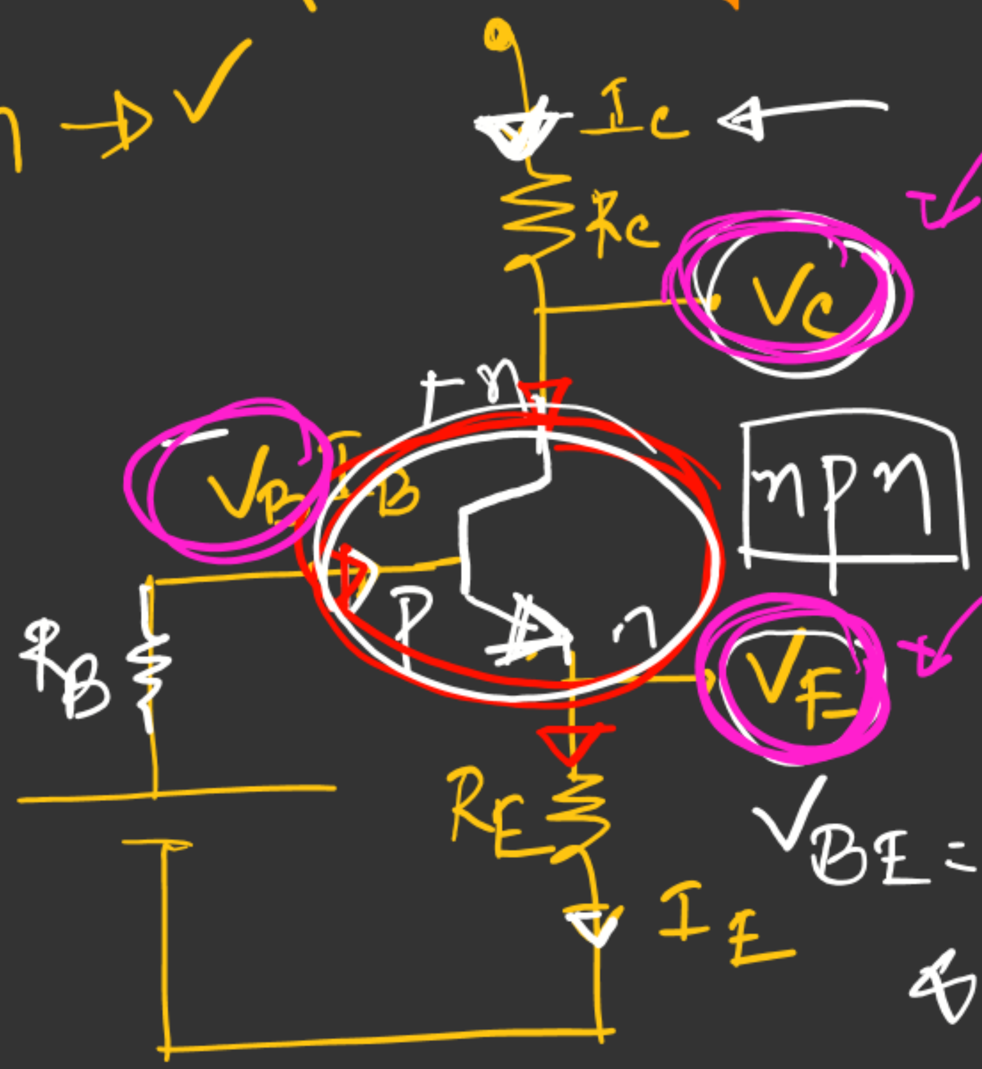
$V_{BE} = 0.7 \rightarrow npn$

$V_{EB} = -0.7\,V$



pnp → X

npn → ✓



$V_{BE} = 0.7\,V$

Si

$$\boxed{I_C + I_B = I_E} \rightarrow$$

$$I_E = I_C + I_B$$

npn/pnp

⊛ $I_C + I_B = I_E$ //

$\downarrow$

$\beta I_B + I_B = I_E$

$\Rightarrow I_E = I_B(1+\beta)$ //

* Current gain,

$$\alpha = \frac{I_C}{I_E}$$

$$\beta = \frac{I_C}{I_B} \Rightarrow I_C = \beta I_B$$

$$\gamma = \frac{I_E}{I_B}$$

⊛ $\alpha = \dfrac{\beta}{\beta+1} = \dfrac{\gamma-1}{\gamma}$

⊛ $P = V_{CE} I_C \leftarrow$ absorbs

* DC load line $\longrightarrow$

$I_C = 0$

$V_{CE}(\text{cut off}) = V_{CC}$

$V_{CE} = 0 \leftarrow$

$$I_C(\text{sat}) = \frac{V_{CC}}{R_C + R_E}$$

Transistor Mode :-

for npn $\rightarrow$ $V_C > V_E$ [active]

$V_E > V_C$ [off]

$V_C = V_E$ [saturation].

For pnp $\rightarrow$ $V_E > V_C$ [active]

$V_C > V_E$ [off]

$V_E = V_C$ [saturation]

$\hookrightarrow$

$V_{CC} - I_C R_C + V_C - V_E + V_{CE} - I_E R_E = 0$

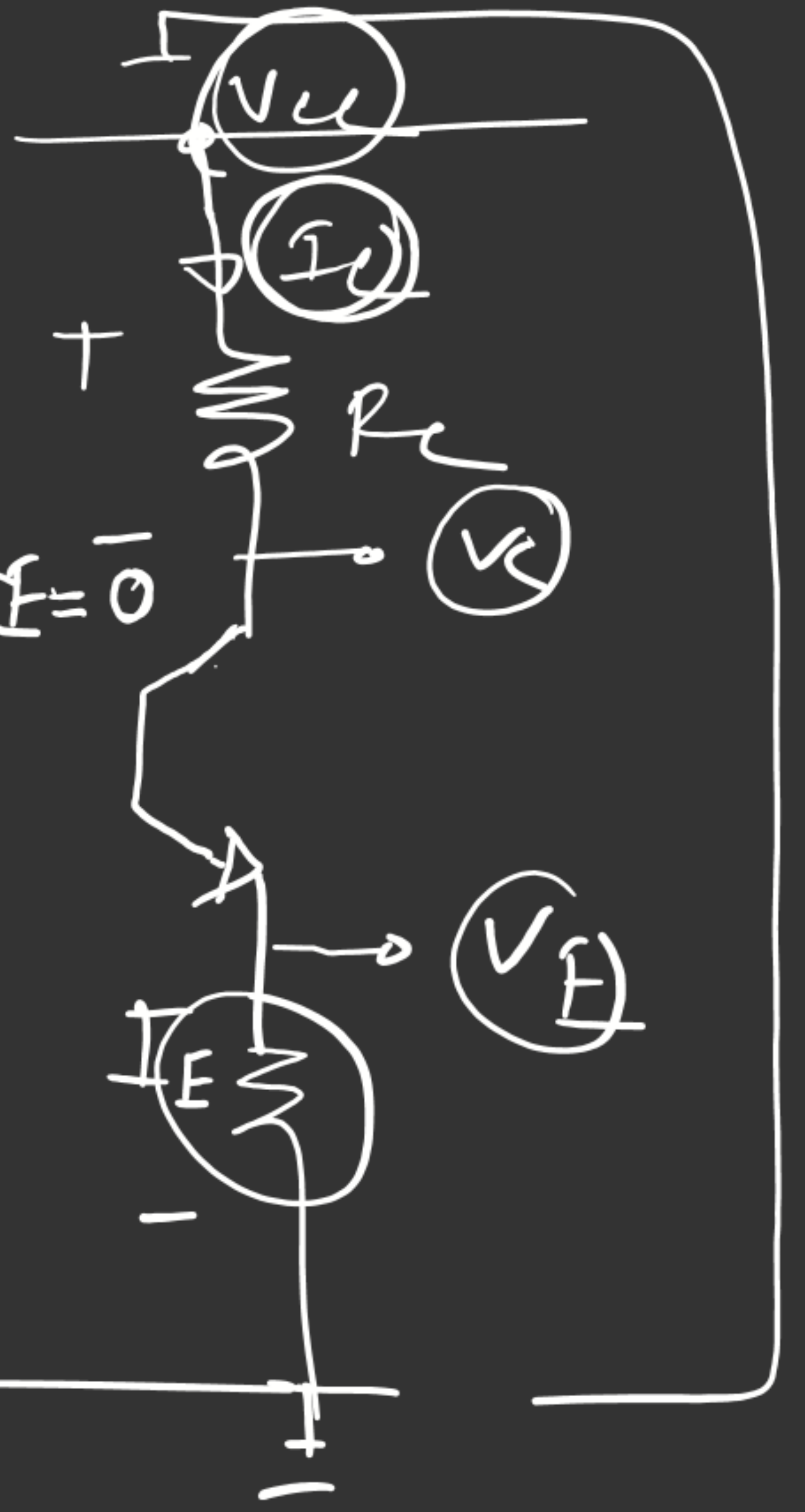
$V_{CE} \downarrow 0$

I_C

$V_{CC} = I_C R_C + I_E R_E$

$= I_C (R_C + R_E)$

$$I_{C(sat)} = \frac{V_{CC}}{R_C + R_E}$$



$V_C = V_E$

$V_C - V_E$

$12 - 12$

$= 0$